

Math 421

Wednesday, November 5

Announcements

- Grace's Drop-In Hours
 - **Today** 12-1 pm VV 311
 - Thursday 9-11 am VV B205
- MLC
 - Proof Table: M-Th 3-7:30 VV B227
 - Course Assistant: Th 4-7 Table
- Homework 8 Due Friday
- Exam 2 – 11/12 5:45-7:15 pm
 - Email me & fill out the form if you have a conflict and need to take the makeup.
 - Sign up for McBurney exam if applicable.

Functional Limits – via Sequences

Def. Let $f: S \rightarrow \mathbb{R}$ and let a be the limit of some sequence in S .

Then $\lim_{x \rightarrow a} f(x) = L$ if for every sequence (x_n) in S satisfying $x_n \neq a$ for all $n \in \mathbb{N}$ and $\lim x_n = a$, we have $\lim f(x_n) = L$.

Example: Let $f(x) = x^2$, $\lim_{x \rightarrow 0} x^2 = 0$.

Activity

Prove carefully using sequences, and their algebraic limit laws, that

$$\lim_{x \rightarrow 3} x^2 - x = 6.$$

Proof.

Functional Limits – Two Perspectives

Theorem. Let $f: S \rightarrow \mathbb{R}$ and let a be the limit of some sequence in S . Then the following statements are equivalent:

1. $\lim_{x \rightarrow a} f(x) = L$
2. for all $\epsilon > 0$ there exists some $\delta > 0$ such that:
if $0 < |x - a| < \delta$ and $x \in S$, then $|f(x) - L| < \epsilon$.
3. for every sequence (x_n) in S satisfying $x_n \neq a$ for all $n \in \mathbb{N}$ and $\lim x_n = a$, we have $\lim f(x_n) = L$.

Proof of Equivalence

$(\epsilon - \delta \Rightarrow \text{Sequences})$:

Proof of Equivalence

($\epsilon - \delta \Leftarrow$ Sequences):

Algebraic Limit Laws

Theorem. Let f and g be functions defined on a domain S and let a be the limit of some sequence in S . If the limits $L = \lim_{x \rightarrow a} f(x)$ and $M = \lim_{x \rightarrow a} g(x)$ exist and are finite. Then

- i. $\lim_{x \rightarrow a} kf(x)$ exists and equals kL for all $k \in \mathbb{R}$.
- ii. $\lim_{x \rightarrow a} (f + g)(x)$ exists and equals $L + M$.
- iii. $\lim_{x \rightarrow a} (fg)(x)$ exists and equals LM .
- iv. $\lim_{x \rightarrow a} (f/g)(x)$ exists and equals L/M provided $M \neq 0$ and $g(x) \neq 0$ for $x \in \text{dom}(g)$.

Activity – Divide & Conquer the proofs of i, ii, and iii.

Discussion Questions/Activity

1. What would you need to do to prove that $\lim_{x \rightarrow a} f(x)$ does not exist?
 - a) According to the $\epsilon - \delta$ definition of the limit
 - b) According to the sequence definition of the limit
2. Show that $\lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right)$ does not exist, using 1.b).

One-Sided & Infinite Limits

- $\lim_{x \rightarrow a^+} f(x) = L$

- $\lim_{x \rightarrow \infty} f(x) = L$

- $\lim_{x \rightarrow a} f(x) = \infty$

Activity

Consider each of the following limits, fill in the following blanks for each:

for each _____ there exists _____ such that

_____ implies _____

1. $\lim_{x \rightarrow a^-} f(x) = \infty$

2. $\lim_{x \rightarrow -\infty} f(x) = -\infty$

Chapter 3: Continuity

Section 17: Continuous Functions

Continuity

Theorem. Let $f: S \rightarrow \mathbb{R}$ and let $x_0 \in S$. Then the following are equivalent:

- a) f is continuous at x_0 ,
- b) Whenever $(x_n) \subseteq S$ and $\lim x_n = x_0$, then $\lim f(x_n) = f(x_0)$,
- c) For all $\epsilon > 0$, there exists a $\delta > 0$ such that: if $|x - x_0| < \delta$ and $x \in S$, then $|f(x) - f(x_0)| < \epsilon$,
- d) $\lim_{x \rightarrow x_0} f(x) = f(x_0)$.

Example

1. Recall we proved $\lim_{x \rightarrow 1} x^2 + x + 1 = 3$. Is $x^2 + x + 1$ continuous at 1?
2. Recall we proved $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = 0$. Define function $g(x)$ so that $g(x) = x^2 \sin\left(\frac{1}{x}\right)$ for all $x \neq 0$ and $g(x)$ is continuous at all $x \in \mathbb{R}$.

Discontinuity

Activity

Prove, using the $\epsilon - \delta$ definition, the following basic properties of continuous functions

1. If $a \in \mathbb{R}$ and $g(x) = a$ is the constant function, then g is continuous at all $c \in \mathbb{R}$.
2. If $f(x) = x$, then f is continuous at all $c \in \mathbb{R}$.

Algebraic Continuity Properties

Theorem 17.3 & 17.4. Assume $f: A \rightarrow \mathbb{R}$ and $g: A \rightarrow \mathbb{R}$ are both continuous at a point $x_0 \in A$. Then

1. $|f(x)|$ is continuous at x_0 ,
2. $kf(x)$ is continuous at x_0 for all $k \in \mathbb{R}$,
3. $f(x) + g(x)$ is continuous at x_0 ,
4. $f(x)g(x)$ is continuous at x_0 , and
5. If $g(x_0) \neq 0$, then $\frac{f(x)}{g(x)}$ is continuous at x_0 .

Example

1. If P is a polynomial, then P is continuous at all $c \in \mathbb{R}$
2. Define $f: [0, \infty) \rightarrow [0, \infty)$ by $f(x) = \sqrt{x}$. Then f is continuous at all $c \in [0, \infty)$.